

# 1 Group Representations, Irreducibility, Characters (Section 9)

**Definition 1.1** (9.1.1). Let  $V$  be a vector space over  $\mathbb{C}$ . Put

$$\mathrm{GL}(V) := \{\mathbb{C}\text{-linear isomorphisms } \phi : V \xrightarrow{\sim} V\}.$$

This is a group under composition.

**Definition 1.2** (9.1.2). Let  $G$  be a group and  $V$  a vector space over  $\mathbb{C}$ . A linear representation of  $G$  on  $V$  is a homomorphism  $\rho : G \rightarrow \mathrm{GL}(V)$ , i.e. for all  $g, h \in G$ ,

$$\rho(gh) = \rho(g)\rho(h).$$

- Example 1.3** (9.1.3). 1. 1-dimensional representations correspond to homomorphisms  $\rho : G \rightarrow \mathbb{C}^\times$ .  
2. The dihedral group  $D_{2n}$  acting on the plane gives a representation.  
3. The regular representation:  $V = \mathbb{C}[G] = \{\sum_{g \in G} a_g[g]\}$  with left translation.  
4. If  $G$  acts on a set  $X$ , then  $\mathbb{C}[X]$  is a representation.

**Definition 1.4** (9.1.4). If  $(V, \rho)$  and  $(V', \rho')$  are representations, a  $G$ -equivariant linear map  $\phi : V \rightarrow V'$  is called a homomorphism of representations. If  $\phi$  is a bijection, it is an isomorphism.

**Definition 1.5** (9.2.1). Let  $\rho : G \rightarrow \mathrm{GL}(V)$  be a linear representation. A subrepresentation is a  $\mathbb{C}$ -linear subspace  $W \subseteq V$  that is stable under  $G$ -action:  $\forall g \in G, \rho(g)(W) \subseteq W$ .

**Example 1.6** (9.2.2). In the regular representation,  $W = \{a \sum_{h \in G} [h] \mid a \in \mathbb{C}\}$  is a 1-dimensional subrepresentation (the trivial representation).

**Example 1.7** (9.2.3). If  $\phi : V \rightarrow W$  is a homomorphism of representations, then  $\ker \phi$  is a subrepresentation of  $V$ .

**Definition 1.8** (9.2.4). Let  $(\rho, W)$  and  $(\rho', W')$  be representations. Their direct sum  $\rho \oplus \rho'$  is defined by  $(\rho \oplus \rho')(g)(w, w') = (\rho(g)w, \rho'(g)w')$ .

**Theorem 1.9** (9.2.5). Let  $G$  be a finite group. If  $W \subseteq V$  is a subrepresentation, then there exists a complementary subrepresentation  $W^\circ \subseteq V$  such that  $V = W \oplus W^\circ$ .

**Construction 1.10** (9.2.6 (Averaging)). Let  $(\rho_1, W_1)$  and  $(\rho_2, W_2)$  be representations of a finite group  $G$ . For any  $\mathbb{C}$ -linear map  $\phi : W_1 \rightarrow W_2$ , define

$$\tilde{\phi} := \frac{1}{|G|} \sum_{g \in G} \rho_2(g) \circ \phi \circ \rho_1(g)^{-1}.$$

Then  $\tilde{\phi}$  is a  $G$ -homomorphism.

**Definition 1.11** (9.2.8). A linear representation  $V$  is called irreducible if the only subrepresentations are  $\{0\}$  and  $V$ .

**Corollary 1.12** (9.2.9). If  $G$  is finite, then every finite dimensional representation  $V$  is completely reducible:  $V \cong W_1 \oplus \cdots \oplus W_k$  with each  $W_i$  irreducible.

**Definition 1.13** (9.3.2). If  $(\rho_1, V)$  and  $(\rho_2, W)$  are representations of  $G$ , their tensor product  $V \otimes W$  has  $G$ -action  $\rho(g)(v \otimes w) = \rho_1(g)v \otimes \rho_2(g)w$ .

**Definition 1.14** (9.3.4). If  $(\rho, V)$  is a representation, the dual representation  $V^* = \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C})$  has action  $(\rho^*(g)(v^*))(v) = v^*(\rho(g^{-1})v)$ .

## 2 Schur's Lemma, Orthogonality Relations (Section 10)

**Definition 2.1** (10.1.2). For a representation  $\rho : G \rightarrow \mathrm{GL}(V)$ , its character is  $\chi_\rho(g) = \mathrm{Tr}(\rho(g))$ .

- Property 2.2** (10.1.3). 1.  $\chi_\rho(1) = \dim V$ .  
2.  $\chi_\rho(hgh^{-1}) = \chi_\rho(g)$ .  
3.  $\chi_{\rho^*}(g) = \chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$ .

**Definition 2.3** (10.1.4). A function  $f : G \rightarrow \mathbb{C}$  is a class function if  $f(hgh^{-1}) = f(h)$  for all  $g, h \in G$ .

**Proposition 2.4** (10.1.6). For representations  $\rho_1, \rho_2$ ,

$$\chi_{\rho_1 \oplus \rho_2}(g) = \chi_{\rho_1}(g) + \chi_{\rho_2}(g), \quad \chi_{\rho_1 \otimes \rho_2}(g) = \chi_{\rho_1}(g) \cdot \chi_{\rho_2}(g).$$

**Definition 2.5** (10.2.1). For two functions  $\chi, \eta$  on  $G$ , define

$$\langle \chi, \eta \rangle = \frac{1}{|G|} \sum_{g \in G} \chi(g) \eta(g^{-1}).$$

**Proposition 2.6** (10.2.4 (Schur's Lemma)). Let  $(\rho_1, V_1)$  and  $(\rho_2, V_2)$  be irreducible  $\mathbb{C}$ -representations of  $G$  and  $\phi : V_1 \rightarrow V_2$  a homomorphism.

1. If  $(\rho_1, V_1)$  and  $(\rho_2, V_2)$  are not isomorphic, then  $\phi = 0$ .
2. If  $(\rho_1, V_1) = (\rho_2, V_2)$ , then  $\phi = \lambda \cdot \mathrm{id}_V$  for some  $\lambda \in \mathbb{C}$ .

**Theorem 2.7** (10.2.3 (Schur Orthogonality)). If  $(\rho_1, V_1)$  and  $(\rho_2, V_2)$  are irreducible representations of a finite group  $G$ , then

$$\langle \chi_{V_1}, \chi_{V_2} \rangle = \begin{cases} 1 & \text{if } \rho_1 \cong \rho_2, \\ 0 & \text{otherwise.} \end{cases}$$

**Proposition 2.8** (10.3.1). If  $V = \bigoplus_i W_i^{\oplus m_i}$  with  $W_i$  irreducible and pairwise non-isomorphic, then

$$\chi_V = \sum_i m_i \chi_{W_i}, \quad m_i = \langle \chi_V, \chi_{W_i} \rangle.$$

**Corollary 2.9** (10.3.2). Two representations with the same character are isomorphic.

**Corollary 2.10** (10.3.3).  $V$  is irreducible iff  $\langle \chi_V, \chi_V \rangle = 1$ .

**Theorem 2.11** (10.5.1). The characters of irreducible representations form an orthonormal basis of the space of class functions. In particular, the number of irreducible representations equals the number of conjugacy classes.

## 3 Rings, Ideals, Quotient Rings (Section 11)

**Definition 3.1** (11.1.1). A ring  $R$  is a set with two binary operations  $+$  and  $\cdot$  such that  $(R, +)$  is an abelian group, multiplication is associative, distributive laws hold, and there exists  $1 \neq 0$  with  $1 \cdot a = a \cdot 1 = a$  for all  $a \in R$ .

**Definition 3.2** (11.1.2). A division ring is a ring where every nonzero element has a multiplicative inverse. A commutative division ring is a field.

**Example 3.3** (11.1.3).  $\mathbb{Z}, \mathbb{Z}_n, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ , polynomial rings, matrix rings, Hamilton quaternions  $\mathbb{H}$ , group rings  $R[G]$ .

**Definition 3.4** (11.1.4). A ring homomorphism  $\phi : R \rightarrow S$  satisfies  $\phi(a + b) = \phi(a) + \phi(b)$ ,  $\phi(ab) = \phi(a)\phi(b)$ , and  $\phi(1) = 1$ . The kernel is  $\ker \phi = \phi^{-1}(0)$ .

- Definition 3.5** (11.1.7). 1. A nonzero  $a \in R$  is a zero-divisor if  $\exists b \neq 0$  with  $ab = 0$  or  $ba = 0$ .  
2.  $u \in R$  is a unit if  $\exists v$  with  $uv = vu = 1$ . The set of units is  $R^\times$ .  
3. A commutative ring with no zero-divisors is an integral domain.

**Lemma 3.6** (11.1.9). *A finite integral domain is a field.*

**Definition 3.7** (11.1.10). *For an integral domain  $R$ , its fraction field  $\text{Frac}(R)$  is the set of equivalence classes  $(a, b)$  with  $b \neq 0$  modulo  $(a, b) \sim (c, d)$  iff  $ad = bc$ .*

**Definition 3.8** (11.2.1). *A subset  $I \subseteq R$  is a left ideal if it is a subgroup under addition and  $RI \subseteq I$ ; similarly for right ideal. A two-sided ideal is an ideal.  $I$  is proper if  $I \neq R$ .*

**Definition 3.9** (11.2.3). *If  $I$  is a two-sided ideal, the quotient ring  $R/I$  has elements  $x + I$  with operations  $(x + I) + (y + I) = (x + y) + I$ ,  $(x + I)(y + I) = xy + I$ .*

**Theorem 3.10** (11.2.4 (Isomorphism Theorems for Rings)). *1. If  $\phi : R \rightarrow S$  is a ring homomorphism, then  $\ker \phi$  is an ideal,  $\phi(R)$  is a subring, and  $R/\ker \phi \cong \phi(R)$ .  
2. If  $I \subseteq J$  are proper ideals, then  $J/I$  is an ideal of  $R/I$  and  $(R/I)/(J/I) \cong R/J$ .  
3. (Lattice isomorphism) There is a bijection between ideals of  $R$  containing  $I$  and ideals of  $R/I$ , preserving inclusion, sums, intersections, and quotients.*

**notation**[11.2.6] For a commutative ring  $R$  and  $a_1, \dots, a_s \in R$ ,  $(a_1, \dots, a_s) = \{\sum x_i a_i \mid x_i \in R\}$  is the ideal generated by them.

**Definition 3.11** (11.2.8). *For ideals  $I, J$ , define  $I + J = \{a + b \mid a \in I, b \in J\}$ , and  $IJ = \{\text{finite sums } \sum a_i b_i \mid a_i \in I, b_i \in J\}$ .*

## 4 Chinese Remainder Theorem, Euclidean Domains, PIDs (Section 12)

**Definition 4.1** (12.1.1). *Two ideals  $I, J$  in a commutative ring are comaximal if  $I + J = R$ .*

**Theorem 4.2** (12.1.2 (Chinese Remainder Theorem)). *Let  $I_1, \dots, I_k$  be ideals in a commutative ring  $R$ . The natural map  $R \rightarrow \prod R/I_i$  has kernel  $\bigcap I_i$ . If the  $I_i$  are pairwise comaximal, then the map is surjective and  $\bigcap I_i = I_1 \cdots I_k$ , so*

$$R/(I_1 \cdots I_k) \cong R/I_1 \times \cdots \times R/I_k.$$

**Definition 4.3** (12.4.1). *An ideal  $\mathfrak{m} \subsetneq R$  is maximal if the only ideals containing  $\mathfrak{m}$  are  $\mathfrak{m}$  and  $R$ .*

**Proposition 4.4** (12.4.3). *In a commutative ring  $R$ , an ideal  $\mathfrak{m}$  is maximal iff  $R/\mathfrak{m}$  is a field.*

**Definition 4.5** (12.5.1). *A proper ideal  $\mathfrak{p} \subsetneq R$  is prime if  $ab \in \mathfrak{p}$  implies  $a \in \mathfrak{p}$  or  $b \in \mathfrak{p}$ .*

**Proposition 4.6** (12.5.3).  *$\mathfrak{p}$  is prime iff  $R/\mathfrak{p}$  is an integral domain.*

**Corollary 4.7** (12.5.4). *Maximal ideals are prime.*

**Definition 4.8** (12.6.1). *A principal ideal domain (PID) is an integral domain in which every ideal is principal.*

**Proposition 4.9** (12.6.3). *Every nonzero prime ideal in a PID is maximal.*

## 5 Unique Factorization Domains (Section 13)

**Definition 5.1** (13.1.1). *An integral domain  $R$  is a Euclidean domain if there exists a norm  $\text{Nm} : R \rightarrow \mathbb{Z}_{\geq 0}$  with  $\text{Nm}(0) = 0$  such that for any  $a, b \neq 0$ , there exist  $q, r \in R$  with  $a = bq + r$  and either  $r = 0$  or  $\text{Nm}(r) < \text{Nm}(b)$ .*

**Proposition 5.2** (13.1.4). *Every Euclidean domain is a PID.*

**Definition 5.3** (13.2.1). *1. For  $a, b \in R$ ,  $a \mid b$  if  $b = ac$  for some  $c$ .  
2. A nonzero  $p \in R$  is prime if  $(p)$  is a prime ideal.  
3. A nonzero non-unit  $r$  is irreducible if  $r = ab$  implies  $a$  or  $b$  is a unit.  
4.  $a, b$  are associated if  $a = bu$  for a unit  $u$ .*

**Proposition 5.4** (13.2.2). *1. Prime elements are irreducible.  
2. In a PID, irreducible elements are prime.*

**Definition 5.5** (13.2.3). *A unique factorization domain (UFD) is an integral domain where every nonzero non-unit is a product of irreducibles, and this factorization is unique up to order and associates.*

**Proposition 5.6** (13.2.6). *In a UFD, an element is prime iff it is irreducible.*

**Proposition 5.7** (13.2.7). *In a UFD, gcd exists and is unique up to associates.*

**Theorem 5.8** (13.3.1). *Every PID is a UFD.*

## 6 UFD Properties of Polynomial Rings (Section 14)

**Theorem 6.1** (14.1.1). *An integral domain  $R$  is a UFD iff  $R[x]$  is a UFD.*

**Corollary 6.2** (14.1.2). *If  $R$  is a UFD, then  $R[x_1, \dots, x_n]$  is a UFD.*

**Lemma 6.3** (14.2.1 (Rational Root Test)). *If  $p(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$  and  $p(r/s) = 0$  with  $\gcd(r, s) = 1$ , then  $r \mid a_0$  and  $s \mid a_n$ .*

**Proposition 6.4** (14.2.3 (Eisenstein's Criterion)). *Let  $\mathfrak{p}$  be a prime ideal in an integral domain  $R$ , and  $f(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0 \in R[x]$ . If  $c_i \in \mathfrak{p}$  for all  $i$  and  $c_0 \notin \mathfrak{p}^2$ , then  $f$  is irreducible.*

**Lemma 6.5** (14.3.1). *Let  $F$  be a field. A polynomial  $f(x) \in F[x]$  is irreducible iff  $F[x]/(f(x))$  is a field.*

**Lemma 6.6** (14.3.3). *If  $f(x) = p_1(x)^{n_1} \cdots p_r(x)^{n_r}$  in  $F[x]$ , then*

$$F[x]/(f(x)) \cong \prod_{i=1}^r F[x]/(p_i(x)^{n_i}).$$

**Corollary 6.7** (14.3.5). *If  $F$  is a field, any finite subgroup of  $F^\times$  is cyclic. In particular, finite fields have cyclic multiplicative groups.*

## 7 Modules and Classification over PIDs (Section 15)

**Definition 7.1** (15.1.1). *Let  $R$  be a ring. A left  $R$ -module is an abelian group  $M$  with an action  $R \times M \rightarrow M$  satisfying  $1 \cdot m = m$ ,  $(r + s) \cdot m = r \cdot m + s \cdot m$ ,  $r \cdot (m + n) = r \cdot m + r \cdot n$ , and  $r \cdot (s \cdot m) = (rs) \cdot m$ .*

**Definition 7.2** (15.1.4). *An  $R$ -module homomorphism  $\phi : M \rightarrow N$  is additive and  $R$ -linear.  $\ker \phi$  and  $\text{Im } \phi$  are submodules.*

**Definition 7.3** (15.2.1). *Let  $M$  be an  $R$ -module and  $X \subseteq M$ . The submodule generated by  $X$  is  $RX = \{\sum a_i x_i\}$ . If  $X$  is finite,  $M$  is finitely generated. If  $M = Rx$ , it is cyclic.  $M$  is free of rank  $n$  if  $M \cong R^{\oplus n}$ .*

**Lemma 7.4** (15.3.1). *If  $R$  is an integral domain and  $N$  is a free  $R$ -module of rank  $n$ , then any  $n + 1$  elements of  $N$  are linearly dependent.*

**Theorem 7.5** (15.3.2 (Structure of submodules of free modules over a PID)). *Let  $R$  be a PID,  $N$  a free  $R$ -module of rank  $n$ , and  $L$  a submodule. Then*

1.  $L$  is free of rank  $\ell \leq n$ .
2. *There exists a basis  $y_1, \dots, y_n$  of  $N$  and nonzero  $a_1, \dots, a_\ell \in R$  with  $a_1 \mid a_2 \mid \dots \mid a_\ell$  such that  $a_1 y_1, \dots, a_\ell y_\ell$  is a basis of  $L$ .*

**Theorem 7.6** (15.3.3 (Fundamental Theorem of Finitely Generated Modules over a PID)). *Let  $R$  be a PID and  $M$  a finitely generated  $R$ -module. Then*

$$M \cong R^{\oplus r} \oplus R/(a_1) \oplus \dots \oplus R/(a_m)$$

*with  $a_1 \mid a_2 \mid \dots \mid a_m$ , and  $r$  and the  $a_i$  are unique up to associates.*

## 8 Field Extensions (Section 16)

**Definition 8.1** (16.1.1). *The characteristic  $\text{char}(F)$  of a field  $F$  is the smallest positive integer  $p$  such that  $p \cdot 1 = 0$ , or 0 if no such integer exists.*

**Definition 8.2** (16.1.3). *The prime field of  $F$  is the smallest subfield containing  $1_F$ ; it is  $\mathbb{F}_p$  if  $\text{char}(F) = p > 0$ , or  $\mathbb{Q}$  if  $\text{char}(F) = 0$ .*

**Definition 8.3** (16.2.2). *The degree of a field extension  $K/F$  is  $[K : F] = \dim_F K$  (as an  $F$ -vector space). The extension is finite if this dimension is finite.*

**Theorem 8.4** (16.2.3 (Tower Law)). *If  $F \subseteq E \subseteq K$  are field extensions, then  $[K : F] = [K : E][E : F]$ .*

**Lemma 8.5** (16.3.1). *Any field homomorphism  $\phi : F \rightarrow E$  is injective.*

**Lemma 8.6** (16.3.3). *If  $p(x) \in F[x]$  is irreducible, then in  $K = F[x]/(p(x))$ , the element  $\theta = x \bmod (p(x))$  satisfies  $p(\theta) = 0$ .*

**Definition 8.7** (16.3.5). *Let  $K/F$  be an extension. The subfield generated by  $\alpha_1, \dots, \alpha_n \in K$  is  $F(\alpha_1, \dots, \alpha_n)$ . If  $K = F(\alpha)$ , it is a simple extension.*

**Theorem 8.8** (16.3.6). *Let  $K/F$  be an extension and  $\alpha \in K$ . Either*

1.  $1, \alpha, \alpha^2, \dots$  are linearly independent over  $F$  (transcendental), then  $F(\alpha) \cong F(x)$ ; or
2. *they are linearly dependent (algebraic), then there is a unique monic irreducible polynomial  $m_{\alpha, F}(x) \in F[x]$  (the minimal polynomial) with  $m_{\alpha, F}(\alpha) = 0$ , and  $F(\alpha) \cong F[x]/(m_{\alpha, F}(x))$ ,  $[F(\alpha) : F] = \deg m_{\alpha, F}$ .*

**Theorem 8.9** (16.4.2). *For a field extension  $K/F$ , the following are equivalent:*

1.  $K/F$  is finite.
2.  $K$  is finitely generated and algebraic over  $F$ .

**Lemma 8.10** (16.4.4). *If  $K/E/F$  are extensions and  $\alpha \in K$ , then  $m_{\alpha, E}(x) \mid m_{\alpha, F}(x)$  in  $E[x]$ .*

**Corollary 8.11** (16.4.5).  $[E(\alpha) : E] \leq [F(\alpha) : F]$ .

**Definition 8.12** (16.4.6). *The composite of intermediate fields  $K_1, K_2$  is  $K_1 K_2$ , the smallest field containing both.*

**Corollary 8.13** (16.4.8). *If  $K_1, K_2$  are finite over  $F$ , then  $[K_1 K_2 : F] \leq [K_1 : F][K_2 : F]$ .*

**Corollary 8.14** (16.4.10). *If  $\alpha, \beta$  are algebraic over  $F$ , then  $\alpha \pm \beta$ ,  $\alpha\beta$ , and  $\alpha/\beta$  (if  $\beta \neq 0$ ) are algebraic over  $F$ .*

**Definition 8.15** (16.4.11). *The algebraic closure of  $F$  in  $K$  is  $\{\alpha \in K \mid \alpha \text{ algebraic over } F\}$ , which is a subfield.*

**Theorem 8.16** (16.4.13). *If  $L/K$  and  $K/F$  are algebraic, then  $L/F$  is algebraic.*

## 9 Normal Extensions (Section 17)

**Definition 9.1** (17.1.1). *A splitting field of  $f(x) \in F[x]$  is an extension  $K/F$  such that  $f$  splits completely in  $K$  and  $K$  is generated by the roots of  $f$ .*

**Theorem 9.2** (17.1.3). *For any field  $F$  and  $f(x) \in F[x]$  of degree  $n$ , a splitting field exists and  $[K : F] \leq n!$ .*

**Lemma 9.3** (17.2.1). *If  $\eta : F \cong F'$  is an isomorphism and  $p(x) \in F[x]$  irreducible, then  $\eta$  induces an isomorphism  $F[x]/(p(x)) \cong F'[x]/(\eta(p)(x))$ .*

**Proposition 9.4** (17.2.3). *Let  $\eta : F \cong F'$  and  $f(x) \in F[x]$ ,  $f'(x) = \eta(f)(x)$ . If  $E$  is a splitting field of  $f$  over  $F$  and  $E'$  a splitting field of  $f'$  over  $F'$ , then  $\eta$  extends to an isomorphism  $E \cong E'$ .*

**Definition 9.5** (17.3.1). *An algebraic extension  $K/F$  is normal if every irreducible polynomial in  $F[x]$  that has a root in  $K$  splits completely in  $K$ .*

**Theorem 9.6** (17.3.2). *A finite extension  $K/F$  is normal iff it is the splitting field of some polynomial in  $F[x]$ .*

**Corollary 9.7** (17.3.3). *If  $K/F$  is finite and normal, then for any intermediate field  $E$ ,  $K/E$  is normal.*

**Definition 9.8** (17.3.4). *A normal closure of a finite extension  $K/F$  is a minimal normal extension of  $F$  containing  $K$ .*

**Lemma 9.9** (17.3.5). *A normal closure exists and is unique up to isomorphism.*

**Definition 9.10** (17.4.1). *In characteristic  $p > 0$ , the Frobenius endomorphism  $\sigma_F : F \rightarrow F$  is  $\sigma(x) = x^p$ .  $F$  is perfect if  $\sigma_F$  is an isomorphism.*

**Proposition 9.11** (17.4.3). *Algebraic extensions of perfect fields are perfect.*

## 10 Separable Extensions and Finite Fields (Section 18)

**Definition 10.1** (18.1.1). *For  $f(x) = a_0 + \dots + a_n x^n \in F[x]$ , the formal derivative is  $D(f)(x) = a_1 + 2a_2 x + \dots + na_n x^{n-1}$ .*

**Theorem 10.2** (18.1.2).  *$f(x)$  has no repeated roots in its splitting field iff  $(f(x), D(f)(x)) = 1$  in  $F[x]$ .*

**Corollary 10.3** (18.1.3). *If  $f$  is irreducible, then  $f$  has repeated roots iff  $D(f)(x) = 0$ .*

**Definition 10.4** (18.1.4). *An irreducible polynomial is separable if it has no repeated roots; otherwise it is inseparable.*

**Corollary 10.5** (18.1.5). *In characteristic 0, all irreducible polynomials are separable.*

**Corollary 10.6** (18.1.6). *If  $\text{char}(F) = p > 0$  and  $f$  is inseparable, then  $f(x) = g(x^p)$  for some irreducible  $g$ . This can happen only if  $F$  is imperfect.*

**Corollary 10.7** (18.1.7). *If  $\text{char}(F) = p > 0$  and  $f$  irreducible, then  $f(x) = g(x^{p^e})$  with  $g$  separable irreducible, and  $f$  has  $\deg g$  distinct roots.*

**Definition 10.8** (18.2.1). *An element  $\alpha$  algebraic over  $F$  is separable if its minimal polynomial is separable. An extension  $K/F$  is separable if every element of  $K$  is separable over  $F$ .*

**Lemma 10.9** (18.2.6). *If  $K = F(\alpha)$  and  $m_{\alpha, F}(x) = g(x^{p^e})$  with  $g$  separable irreducible, then*

$$\# \text{Hom}_F(F(\alpha), M) = \deg g \leq [F(\alpha) : F],$$

*with equality iff  $\alpha$  is separable.*

**Corollary 10.10** (18.2.8). *Let  $K/F$  be finite and  $M$  a normal extension containing  $K$ . Then*

$$\# \text{Hom}_F(K, M) \leq [K : F].$$

*Equality holds iff  $K/F$  is separable (equivalently,  $K$  is generated over  $F$  by separable elements).*

**Theorem 10.11** (18.2.3). *1. If  $\alpha$  is separable over  $F$ , then  $F(\alpha)/F$  is separable.  
2. If  $L/K$  and  $K/F$  are separable, then  $L/F$  is separable.*

**Definition 10.12** (18.2.10). *For a finite extension  $K/F$ , define  $K^s = \{\alpha \in K \mid \alpha \text{ separable over } F\}$  (the maximal separable subextension). Then  $[K : F]_{\text{sep}} = [K^s : F]$  and  $[K : F]_{\text{insep}} = [K : K^s]$ .*

**Theorem 10.13** (18.3.1 (Primitive Element Theorem)). *A finite separable extension is generated by one element.*

**Theorem 10.14** (18.4.1). *1. If  $F$  is a finite field, then  $\text{char}(F) = p$  and  $|F| = p^n$  for some  $n$ .  
2. For each prime power  $p^n$ , there exists a field  $\mathbb{F}_{p^n}$  of  $p^n$  elements, unique up to isomorphism; it is the splitting field of  $x^{p^n} - x$  over  $\mathbb{F}_p$ .*

**Lemma 10.15** (18.4.3). *1.  $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$  iff  $m \mid n$ , and the inclusion is unique.  
2.  $\mathbb{F}_{p^n} = \mathbb{F}_p(\alpha)$  for some  $\alpha$  with minimal polynomial of degree  $n$ .*

## Homework 3

**Problem 1.** Prove that every group of order  $5 \cdot 7 \cdot 47$  is cyclic.

*Proof.* Let  $|G| = 5 \cdot 7 \cdot 47$ . By Sylow's Third Theorem, the number of Sylow  $p$ -subgroups, denoted  $n_p$ , satisfies  $n_p \equiv 1 \pmod{p}$  and  $n_p \mid \frac{|G|}{p}$ .

- For  $p = 47$ :  $n_{47} \equiv 1 \pmod{47}$  and  $n_{47} \mid 35$ . The only divisor of 35 that is 1 mod 47 is 1. So  $n_{47} = 1$ .
- For  $p = 7$ :  $n_7 \equiv 1 \pmod{7}$  and  $n_7 \mid 235$ . Divisors of 235 are 1, 5, 47, 235. None of 5, 47, 235 are 1 mod 7. So  $n_7 = 1$ .
- For  $p = 5$ :  $n_5 \equiv 1 \pmod{5}$  and  $n_5 \mid 329$ . Divisors of 329 are 1, 7, 47, 329. None of 7, 47, 329 are 1 mod 5. So  $n_5 = 1$ .

Since all Sylow subgroups are normal and have pairwise coprime orders,  $G$  is isomorphic to their direct product:  $G \cong P_5 \times P_7 \times P_{47}$ . Each Sylow subgroup is of prime

order, hence cyclic. The direct product of cyclic groups of pairwise coprime orders is cyclic. Therefore  $G \cong \mathbb{Z}/5 \times \mathbb{Z}/7 \times \mathbb{Z}/47 \cong \mathbb{Z}/1645$ , which is cyclic.

**Problem 2.** Let  $P \in \text{Syl}_p(G)$  and assume  $N \trianglelefteq G$ . Prove that  $P \cap N$  is a Sylow  $p$ -subgroup of  $N$ . Deduce that  $PN/N$  is a Sylow  $p$ -subgroup of  $G/N$ .

*Proof.* Let  $|G| = p^k m$  with  $\gcd(p, m) = 1$ . Then  $|P| = p^k$ . Let  $|N| = p^a b$  with  $\gcd(p, b) = 1$  and  $a \leq k$ . Then  $P \cap N$  is a  $p$ -subgroup of  $N$ . By Sylow's First Theorem,  $P \cap N$  is contained in some Sylow  $p$ -subgroup  $Q$  of  $N$ ; thus  $|Q| = p^a$ . Since  $Q$  is a  $p$ -subgroup of  $G$ , it is contained in some conjugate of  $P$ ; by Sylow's Second Theorem,  $Q \leq gPg^{-1}$  for some  $g \in G$ . Because  $N \trianglelefteq G$ ,  $g^{-1}Qg \leq N$ . Also  $g^{-1}Qg \leq P$ . Hence  $g^{-1}Qg \leq P \cap N$ . Conjugation preserves order, so  $|P \cap N| \geq p^a$ . But  $P \cap N \leq N$  implies  $|P \cap N| \leq p^a$ . Thus  $|P \cap N| = p^a$ , so  $P \cap N$  is a Sylow  $p$ -subgroup of  $N$ .

By the Second Isomorphism Theorem,  $PN/N \cong P/(P \cap N)$ . Then  $|PN/N| = \frac{|P|}{|P \cap N|} = p^{k-a}$ .

The order of  $G/N$  is  $\frac{|G|}{|N|} = p^{k-a} \frac{m}{b}$  with  $\gcd(p, \frac{m}{b}) = 1$ . Hence  $PN/N$  is a Sylow  $p$ -subgroup of  $G/N$ .

**Problem 3.** Prove that a finite group  $G$  is nilpotent if and only if whenever  $a, b \in G$  with  $\gcd(|a|, |b|) = 1$  then  $ab = ba$ .

*Proof.* ( $\Rightarrow$ ) Suppose  $G$  is nilpotent. A finite nilpotent group is the direct product of its Sylow subgroups:  $G \cong P_{p_1} \times \cdots \times P_{p_k}$ . If  $\gcd(|a|, |b|) = 1$ , then the non-identity coordinates of  $a$  and  $b$  lie in disjoint sets of Sylow factors, and elements from distinct factors commute. Hence  $ab = ba$ .

( $\Leftarrow$ ) Suppose elements of coprime order commute. Let  $P$  be a Sylow  $p$ -subgroup and  $x \in P$ . For any prime  $q \neq p$ , let  $Q$  be a Sylow  $q$ -subgroup and  $y \in Q$ . Since  $|x|$  is a power of  $p$  and  $|y|$  a power of  $q$ ,  $\gcd(|x|, |y|) = 1$ ; by hypothesis  $xy = yx$ . Thus  $P$  commutes pointwise with every other Sylow subgroup. It follows that  $P$  is normalized by the subgroup generated by all Sylow subgroups, which is  $G$  itself. Hence every Sylow subgroup is normal. A finite group with all Sylow subgroups normal is the direct product of them, hence nilpotent.

**Problem 4.** Let  $P$  be a group of order  $p^a$  and  $H$  a normal subgroup of  $P$  of order  $p^b$ . Then for every  $c \in \{0, 1, \dots, b\}$ ,  $H$  contains a subgroup of order  $p^c$  that is normal in  $P$ .

*Proof.* We use induction on  $b$ . The case  $b = 0$  is trivial. Assume the claim holds for normal subgroups of order  $p^k$  with  $k < b$ . Since  $H \trianglelefteq P$  and  $H \neq 1$ , we have  $H \cap Z(P) \neq 1$ . By Cauchy's theorem, there exists

$z \in H \cap Z(P)$  of order  $p$ . Let  $Z_1 = \langle z \rangle$ ; then  $Z_1 \trianglelefteq P$ ,  $Z_1 \leq H$ , and  $|Z_1| = p$ . This gives the case  $c = 1$ .

For  $c > 1$ , consider  $P/Z_1$ . Since  $Z_1 \leq H \trianglelefteq P$ , we have  $H/Z_1 \trianglelefteq P/Z_1$  and  $|H/Z_1| = p^{b-1}$ . By the induction hypothesis,  $H/Z_1$  contains a subgroup  $\bar{K} \trianglelefteq P/Z_1$  of order  $p^{c-1}$ . By the Lattice Isomorphism Theorem,  $\bar{K} = K/Z_1$  for some subgroup  $K \leq H$  containing  $Z_1$ . Then  $|K| = p^{c-1} \cdot p = p^c$  and  $K \trianglelefteq P$ .

**Problem 5.** Let  $\rho : G \rightarrow GL(V)$  be a representation of a finite group  $G$  of dimension  $d$ . Let  $\chi_\rho : G \rightarrow \mathbb{C}$  be its character. Show that for every  $g \in G$ ,  $|\chi_\rho(g)| \leq d$ . Moreover, show that  $\chi_\rho(g) = d$  if and only if  $g \in \ker \rho$ .

*Proof.* Since  $G$  is finite,  $g$  has finite order  $m$ . Then  $\rho(g)^m = I_d$ , so  $\rho(g)$  is diagonalizable and its eigenvalues  $\lambda_1, \dots, \lambda_d$  are roots of unity, hence  $|\lambda_i| = 1$ . Then  $|\chi_\rho(g)| = |\sum \lambda_i| \leq \sum |\lambda_i| = d$ .

If  $\chi_\rho(g) = d$ , then  $\sum \lambda_i = d$ . Because each  $|\lambda_i| = 1$ , this forces  $\lambda_i = 1$  for all  $i$ ; thus  $\rho(g) = I_d$ , i.e.,  $g \in \ker \rho$ .

Conversely, if  $g \in \ker \rho$ , then  $\rho(g) = I_d$  and its trace is  $d$ .

**Problem 6.** Let  $(\rho, V = \mathbb{C}^n)$  be a  $\mathbb{C}$ -representation of  $G$  and let  $z \in Z(G)$  be central.

1. Show that for each  $g \in G$ ,  $\rho(g)$  is diagonalizable in  $GL_n(\mathbb{C})$ .
2. If  $\lambda$  is an eigenvalue of  $\rho(z)$ , then the subspace  $V^{\rho(z)=\lambda} = \{v \in V \mid \rho(z)v = \lambda v\}$  is a subrepresentation.
3. Show that if  $V$  is irreducible, then for every  $z \in Z(G)$ ,  $\rho(z)$  is a scalar matrix.

*Proof.*

1. The element  $g$  has finite order, so  $\rho(g)^m = I_n$ . The polynomial  $x^m - 1$  annihilates  $\rho(g)$  and has no repeated roots; therefore  $\rho(g)$  is diagonalizable.
2. Let  $v \in V^{\rho(z)=\lambda}$  and  $g \in G$ . Since  $z$  is central,  $\rho(z)\rho(g) = \rho(g)\rho(z)$ . Then  $\rho(z)(\rho(g)v) = \rho(g)(\rho(z)v) = \rho(g)(\lambda v) = \lambda(\rho(g)v)$ . Hence  $\rho(g)v$  is also an eigenvector for  $\lambda$ , so the subspace is  $G$ -invariant.
3. For an irreducible  $V$ , the eigenspace  $V^{\rho(z)=\lambda}$  is non-zero (pick any eigenvalue) and is a subrepresentation, so it must equal  $V$ . Thus  $\rho(z)v = \lambda v$  for all  $v \in V$ , i.e.,  $\rho(z) = \lambda I_n$ .

**Problem 7.** Let  $V$  be a  $\mathbb{C}$ -vector space. Then  $V \otimes V = \text{Sym}^2(V) \oplus \Lambda^2(V)$ , where  $\text{Sym}^2(V)$  is spanned by  $v \otimes w + w \otimes v$  and  $\Lambda^2(V)$  by  $v \otimes w - w \otimes v$ .

1. If  $V$  is a representation of  $G$ , show that both  $\text{Sym}^2(V)$  and  $\Lambda^2(V)$  are subrepresentations of  $V \otimes V$ .

2. Show the character identities:

$$\chi_{\text{Sym}^2(V)}(g) = \frac{1}{2}(\chi_V(g)^2 + \chi_V(g^2)), \quad \chi_{\Lambda^2(V)}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2)).$$

*Proof.*

1. Define  $\theta : V \otimes V \rightarrow V \otimes V$  by  $\theta(v \otimes w) = w \otimes v$ . Then  $\theta$  commutes with the  $G$ -action because  $g \cdot \theta(v \otimes w) = gw \otimes gv = \theta(gv \otimes gw) = \theta(g \cdot (v \otimes w))$ . The  $+1$  eigenspace of  $\theta$  is  $\text{Sym}^2(V)$  and the  $-1$  eigenspace is  $\Lambda^2(V)$ ; both are  $G$ -invariant.
2. Choose a basis  $e_1, \dots, e_n$  of eigenvectors of  $g$  with eigenvalues  $\lambda_i$ . Then  $\chi_V(g) = \sum \lambda_i$ ,  $\chi_V(g^2) = \sum \lambda_i^2$ . The eigenvalues on  $\text{Sym}^2(V)$  are  $\lambda_i \lambda_j$  for  $i \leq j$ , and on  $\Lambda^2(V)$  for  $i < j$ . Hence

$$\chi_{\text{Sym}^2(V)}(g) = \sum_{i \leq j} \lambda_i \lambda_j, \quad \chi_{\Lambda^2(V)}(g) = \sum_{i < j} \lambda_i \lambda_j.$$

Now  $(\sum \lambda_i)^2 = \sum \lambda_i^2 + 2 \sum_{i < j} \lambda_i \lambda_j$ . Adding  $\sum \lambda_i^2$  gives the first formula, subtracting gives the second.

**Problem 8.** Let  $G$  act on a finite set  $X$ . This induces a representation on  $V = \mathbb{C}[X] = \{\sum a_x[x] \mid a_x \in \mathbb{C}\}$ . Show that for each  $g \in G$ ,  $\chi_V(g) = |\{x \in X \mid gx = x\}|$ .

*Proof.* The basis  $\{[x]\}_{x \in X}$  is permuted by  $g$ :  $g \cdot [x] = [gx]$ . The matrix of  $g$  in this basis has entry 1 at  $(x, x)$  if and only if  $gx = x$ ; otherwise the diagonal entry is 0. Hence the trace equals the number of fixed points.

**Problem 9.** Let  $H \leq G$  be a finite group,  $(\rho, V)$  a representation of  $H$ . The induced representation  $\text{Ind}_H^G V$  is defined as

$\{f : G \rightarrow V \mid f(hg) = \rho(h)f(g) \ \forall h \in H, g \in G\}$  with  $(g \cdot f)(x) = f(xg)$ .

1. Show that  $\text{Ind}_H^G V$  is finite-dimensional; what is its dimension?
2. Show that  $\chi_{\text{Ind}_H^G V}(g) = \frac{1}{|H|} \sum_{x \in G, x^{-1}gx \in H} \chi_\rho(x^{-1}gx)$ .

*Proof.*

1. Let  $G = \bigsqcup_{i=1}^k Hc_i$  with  $k = [G : H]$ . Any  $f$  is determined by its values on the coset representatives. For each  $i$ , choose any  $v_i \in V$  and set  $f(c_i) = v_i$ ; then extend by  $f(hc_i) = \rho(h)v_i$ . This gives an isomorphism  $\text{Ind}_H^G V \cong V^{\oplus k}$ . Hence  $\dim \text{Ind}_H^G V = [G : H] \dim V$ .
2. Decompose  $\text{Ind}_H^G V = \bigoplus_{i=1}^k W_i$  where  $W_i = \{f \mid f(x) = 0 \text{ for } x \notin Hc_i\}$ . Each  $W_i \cong V$ . For a given  $g$ , the action maps  $W_i$  to  $W_j$  where  $Hc_i g = Hc_j$ . The trace gets contributions only from those  $i$  with  $Hc_i g = Hc_i$ , i.e.,  $c_i g c_i^{-1} \in H$ . On such a block, the action corresponds to  $\rho(c_i g c_i^{-1})$  on  $V$ , whose trace is  $\chi_\rho(c_i g c_i^{-1})$ . Summing over these  $i$  and then averaging over the  $|H|$  elements in each coset yields the given formula.

**Problem 10.** Let  $(\rho_1, V_1), (\rho_2, V_2)$  be representations of  $G$ . Define a representation on  $W = \text{Hom}_{\mathbb{C}}(V_1, V_2)$  by  $(\rho(g)f)(v) = \rho_2(g)(f(\rho_1(g^{-1})v))$ . Show that this gives a representation and its character is  $\chi_{\rho_1^*} \chi_{\rho_2}$ .

*Proof.* One verifies  $\rho(g_1 g_2) = \rho(g_1) \rho(g_2)$ . The natural isomorphism  $\text{Hom}(V_1, V_2) \cong V_1^* \otimes V_2$  intertwines the actions:  $g \cdot (\varphi \otimes v) = (\varphi \circ \rho_1(g)^{-1}) \otimes \rho_2(g)v$ . Hence the character is  $\chi_{V_1^*}(g) \chi_{V_2}(g) = \overline{\chi_{V_1}(g)} \chi_{V_2}(g) = \chi_{\rho_1^*}(g) \chi_{\rho_2}(g)$ .

## Homework 4

**Problem 1.** Consider the ring of Gaussian integers  $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ . Verify that  $\mathbb{Z}[i]/(3+i) \cong \mathbb{Z}/10\mathbb{Z}$ .

*Proof.* Since  $10 = (3+i)(3-i) \in (3+i)$ , we have  $(10) \subseteq (3+i)$ . Define  $\phi : \mathbb{Z} \rightarrow \mathbb{Z}[i]/(3+i)$  by  $\phi(n) = n + (3+i)$ . If  $n \in \ker \phi$ , then  $n = (3+i)(x+yi)$  for some  $x, y \in \mathbb{Z}$ . Expanding gives  $n = (3x - y) + (x + 3y)i$ ; the imaginary part forces  $x = -3y$ , so  $n = 3(-3y) - y = -10y$ . Thus  $\ker \phi = 10\mathbb{Z}$ . Surjectivity: in the quotient,  $i \equiv -3$ , so any  $a + bi \equiv a - 3b \in \mathbb{Z}$ . Hence  $\phi$  is surjective, and the First Isomorphism Theorem gives  $\mathbb{Z}/10\mathbb{Z} \cong \mathbb{Z}[i]/(3+i)$ .

**Problem 2.** Prove that a quotient of a PID by a prime ideal is a PID.

*Proof.* Let  $R$  be a PID and  $P$  a prime ideal. If  $P = (0)$ , then  $R/P \cong R$  is a PID. If  $P \neq (0)$ , then in a PID every nonzero prime ideal is maximal, so  $R/P$  is a field. Fields are trivially PIDs.

**Problem 3.** A commutative ring  $R$  is called a *local ring* if it has a unique maximal ideal. Prove that if  $R$  is a local ring with maximal ideal  $\mathfrak{m}$  then every element of  $R \setminus \mathfrak{m}$  is a unit. Conversely, if  $R$  is commutative and the set of non-units forms an ideal  $I$ , then  $R$  is local with unique maximal ideal  $I$ .

*Proof.* Assume  $R$  is local with maximal ideal  $\mathfrak{m}$ . For any  $x \notin \mathfrak{m}$ , the ideal  $(x)$  is not contained in  $\mathfrak{m}$ ; since every proper ideal is contained in some maximal ideal (here  $\mathfrak{m}$  is the only one),  $(x)$  cannot be proper, so  $(x) = R$  and  $x$  is a unit.

Conversely, let  $I$  be the set of non-units. Then  $1 \notin I$ , so  $I$  is proper. Any proper ideal consists of non-units, hence is contained in  $I$ . Thus  $I$  is maximal and unique.

**Problem 4.** Determine whether the following polynomials are irreducible in the rings indicated. For those reducible, factor into irreducibles.

1.  $x^2 + x + 1$  in  $\mathbb{F}_3[x]$ .
2.  $x^3 + x + 1$  in  $\mathbb{F}_3[x]$ .
3.  $x^4 + 1$  in  $\mathbb{F}_5[x]$ .
4.  $x^4 + 10x^2 + 1$  in  $\mathbb{Z}[x]$ .

*Solution.*

1. Reducible:  $(x+2)^2$ .
2. Reducible:  $(x+2)(x^2+x+2)$ .
3. Reducible:  $(x^2+2)(x^2+3)$ .
4. Irreducible.

**Problem 5.** An integral domain  $R$  where every ideal generated by two elements is principal is called a *Bézout domain*.

1. Prove that  $R$  is a Bézout domain iff every pair  $a, b$  has a gcd  $d$  that can be written as  $d = ax + by$ .
2. Prove that every finitely generated ideal of a Bézout domain is principal.
3. Let  $F$  be the fraction field of  $R$ . Prove that every element of  $F$  can be written as  $a/b$  with  $\gcd(a, b) = 1$ .
4. Prove that  $R$  is a PID iff  $R$  is a UFD and a Bézout domain.

*Proof.*

1.  $(\Rightarrow)$  If  $(a, b) = (d)$ , then  $d = ax + by$  and  $d \mid a, b$ . If  $c \mid a, b$  then  $(a, b) \subseteq (c)$ , so  $d \in (c)$  and  $c \mid d$ .  $(\Leftarrow)$  If a gcd  $d = ax + by$  exists, then  $(d) \subseteq (a, b)$  and clearly  $(a, b) \subseteq (d)$ , hence  $(a, b) = (d)$ .
2. Induction:  $(a_1, \dots, a_n) = ((a_1, \dots, a_{n-1}), a_n) = (d, a_n)$  which is principal by the two-generator property.
3. Write  $\alpha = u/v$ . Let  $d = \gcd(u, v) = ux + vy$ . Write  $u = ad, v = bd$ . Then  $\alpha = a/b$ . If  $c \mid a, b$  then  $cd \mid u, v$  so  $cd \mid d$ ; thus  $c$  is a unit, so  $\gcd(a, b) = 1$ .
4. A PID is a UFD and Bézout. Conversely, let  $R$  be a UFD and Bézout. Take a nonzero ideal  $I$ . Pick  $a_1 \in I$ . If  $(a_1) \neq I$ , choose  $a_2 \in I \setminus (a_1)$ . Then  $(a_1, a_2) = (d_2)$  gives a proper divisor of  $a_1$ . Since factorisation in a UFD is finite, this process terminates, giving  $I = (d_m)$ .

**Problem 6.** Let  $\phi : R \rightarrow S$  be a homomorphism of commutative rings.

1. Prove that if  $P$  is prime in  $S$  then  $\phi^{-1}(P)$  is prime in  $R$ .
2. Prove that if  $M$  is maximal in  $S$  and  $\phi$  is surjective then  $\phi^{-1}(M)$  is maximal in  $R$ . Give an example where surjectivity fails.

*Proof.*

1. If  $ab \in \phi^{-1}(P)$ , then  $\phi(a)\phi(b) \in P$ ; since  $P$  is prime,  $\phi(a) \in P$  or  $\phi(b) \in P$ , so  $a \in \phi^{-1}(P)$  or  $b \in \phi^{-1}(P)$ . Also  $1_R \notin \phi^{-1}(P)$  because  $1_S \notin P$ .

2. By the isomorphism theorems,  $S \cong R/\ker \phi$ . The lattice correspondence gives a bijection between ideals of  $R$  containing  $\ker \phi$  and ideals of  $S$ . Since  $M$  is maximal in  $S$ , its preimage is maximal among ideals containing  $\ker \phi$ ; any ideal strictly containing  $\phi^{-1}(M)$  would map to an ideal strictly containing  $M$ , contradiction. Counterexample: inclusion  $\mathbb{Z} \hookrightarrow \mathbb{Q}$ ; (0) is maximal in  $\mathbb{Q}$  but its preimage (0) is not maximal in  $\mathbb{Z}$ .

**Problem 7.** Let  $F$  be a field and

$f(x) = a_n x^n + \cdots + a_0 \in F[x]$  with  $a_n \neq 0$ . Define the reverse polynomial  $g(x) = x^n f(1/x)$ .

1. Describe the coefficients of  $g$ .
2. If  $a_0 \neq 0$ , prove that  $f$  is irreducible iff  $g$  is irreducible.

*Proof.*

1.  $g(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_n$ .
2. Let  $\text{rev}_d(h)(x) = x^d h(1/x)$ . If  $f = uv$  with  $\deg u = r$ ,  $\deg v = s \geq 1$ , then  $g = \text{rev}_r(u) \text{rev}_s(v)$ . Conversely, if  $g = uv$  nontrivially, then  $f = \text{rev}_n(g) = \text{rev}_r(u) \text{rev}_s(v)$  gives a nontrivial factorisation.

**Problem 8.** Let  $R = \mathbb{Z}[\sqrt{-5}]$ . Define ideals

$$I_2 = (2, 1 + \sqrt{-5}), \quad I_3 = (3, 2 + \sqrt{-5}), \quad I'_3 = (3, 2 - \sqrt{-5}).$$

1. Prove each  $I_2, I_3, I'_3$  is non-principal.
2. Show  $I_2^2 = (2)$ .
3. Show  $I_2 I_3 = (1 - \sqrt{-5})$  and  $I_2 I'_3 = (1 + \sqrt{-5})$ . Deduce (6) =  $I_2^2 I_3 I'_3$ .

*Proof.*

1. Use the norm  $N(a + b\sqrt{-5}) = a^2 + 5b^2$ . For  $I_2$ , if  $I_2 = (\alpha)$  then  $\alpha \mid 2, \alpha \mid 1 + \sqrt{-5}$  so  $N(\alpha) \mid \gcd(4, 6) = 2$ .  $N(\alpha) = 1$  would give  $I_2 = R$ , impossible because the image of  $I_2$  under the map  $a + b\sqrt{-5} \mapsto a + b \pmod{2}$  is 0.  $N(\alpha) = 2$  has no integer solutions. Similar arguments for  $I_3, I'_3$ .
2.  $I_2^2 = (4, 2 + 2\sqrt{-5}, -4 + 2\sqrt{-5})$ . Every generator is divisible by 2, so  $I_2^2 \subseteq (2)$ . Also  $2 = 6 - 4 \in I_2^2$  because  $6 = (2 + 2\sqrt{-5}) - (-4 + 2\sqrt{-5})$  and  $4 \in I_2^2$ . Hence  $(2) \subseteq I_2^2$ .
3. Compute  $I_2 I_3 = (6, 4 + 2\sqrt{-5}, 3 + 3\sqrt{-5}, -3 + 3\sqrt{-5})$ . Each generator is a multiple of  $1 - \sqrt{-5}$ , so  $I_2 I_3 \subseteq (1 - \sqrt{-5})$ . Conversely,  $1 - \sqrt{-5} = (3 + 3\sqrt{-5}) - (4 + 2\sqrt{-5}) \in I_2 I_3$ . Similarly for  $I_2 I'_3$ . Then

$$I_2^2 I_3 I'_3 = (2)(1 - \sqrt{-5})(1 + \sqrt{-5}) = (2)(6) = (6).$$

**Problem 9.** Let  $M$  be an  $R$ -module,  $N$  a submodule.

Prove that if  $M/N$  and  $N$  are finitely generated, then  $M$  is finitely generated.

*Proof.* Let  $\bar{x}_1, \dots, \bar{x}_k$  generate  $M/N$  with lifts  $x_i \in M$ , and let  $y_1, \dots, y_\ell$  generate  $N$ . For any  $m \in M$ , write  $m + N = \sum r_i \bar{x}_i$  so that  $m - \sum r_i x_i \in N$ . Hence  $m = \sum r_i x_i + \sum s_j y_j$ . Thus  $\{x_i, y_j\}$  generate  $M$ .

**Problem 10.** Let  $f : A \rightarrow B$  and  $g : B \rightarrow A$  be  $R$ -module homomorphisms with  $g \circ f = 1_A$ . Show that  $B = \text{Im } f \oplus \ker g$ .

*Proof.* If  $x \in \text{Im } f \cap \ker g$ , then  $x = f(a)$  and  $g(x) = 0$ . Then  $0 = g(f(a)) = a$ , so  $x = 0$ . For any  $b \in B$ , write  $b = f(g(b)) + (b - f(g(b)))$ . Clearly  $f(g(b)) \in \text{Im } f$  and  $g(b - f(g(b))) = g(b) - g(b) = 0$ . Hence  $B = \text{Im } f + \ker g$ , and the sum is direct.

## Homework 5

**Problem 1.**

1. Show that  $x^3 - 2x - 2$  is irreducible over  $\mathbb{Q}$ .
2. Let  $K = \mathbb{Q}[x]/(x^3 - 2x - 2)$  and  $\theta$  the image of  $x$ . Show that  $\{1, \theta, \theta^2\}$  is a  $\mathbb{Q}$ -basis of  $K$ .
3. Compute  $(1 + \theta)(1 + \theta + \theta^2)$  and  $\frac{1+\theta}{1+\theta+\theta^2}$  in  $K$ .

*Solution.*

1. Possible rational roots  $\pm 1, \pm 2$  fail. Since cubic, irreducible.
2. The polynomial is monic irreducible, hence the minimal polynomial of  $\theta$ .
3.  $(1 + \theta)(1 + \theta + \theta^2) = 3 + 4\theta + 2\theta^2$  using  $\theta^3 = 2\theta + 2$ . The inverse of  $1 + \theta + \theta^2$  is  $\frac{5}{3} + \frac{1}{3}\theta - \frac{2}{3}\theta^2$  (solving a linear system). Then  $\frac{1+\theta}{1+\theta+\theta^2} = \frac{1}{3} + \frac{2}{3}\theta - \frac{1}{3}\theta^2$ .

**Problem 2.** Find a basis of the extensions:

1.  $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ .
2.  $\mathbb{Q}(\sqrt{3}, \sqrt{-1}, \omega)$  with  $\omega = \frac{1}{2}(-1 + \sqrt{-3})$ .

*Solution.*

1.  $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ .
2. Note  $\omega = \frac{1}{2}(-1 + \sqrt{-3})$ , so  $\sqrt{-3} = 2\omega + 1$  and  $\sqrt{-1} = i$ . Then  $\mathbb{Q}(\sqrt{3}, i, \omega) = \mathbb{Q}(\sqrt{3}, i)$ . A basis:  $\{1, \sqrt{3}, i, i\sqrt{3}\}$ .

**Problem 3.** Prove that  $K_1 = \mathbb{F}_{11}[x]/(x^2 + 1)$  and  $K_2 = \mathbb{F}_{11}[y]/(y^2 + 2y + 2)$  are fields with 121 elements, and the map sending  $p(\bar{x})$  to  $p(\bar{y} + 1)$  is an isomorphism.

*Proof.*  $-1$  is not a square in  $\mathbb{F}_{11}$ ; discriminant  $4 - 8 = -4 = 7$  not a square. Hence both polynomials irreducible, giving quadratic extensions, so each has  $11^2 = 121$  elements. The map  $\phi(p(x)) = p(y + 1)$  sends  $x^2 + 1$  to  $(y + 1)^2 + 1 = y^2 + 2y + 2 = 0$ , so  $(x^2 + 1) \subseteq \ker \phi$ . By maximality,  $\ker \phi = (x^2 + 1)$ . Hence

$\phi$  induces an injective homomorphism between fields of the same finite cardinality, thus an isomorphism.

**Problem 4.** Prove  $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ . Conclude  $[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = 4$  and find the minimal polynomial.

*Proof.* Let  $\alpha = \sqrt{2} + \sqrt{3}$ . Then  $\alpha^{-1} = \sqrt{3} - \sqrt{2}$ , so  $\sqrt{2} = \frac{\alpha - \alpha^{-1}}{2}$  and  $\sqrt{3} = \frac{\alpha + \alpha^{-1}}{2}$ . Hence both lie in  $\mathbb{Q}(\alpha)$ . Thus equality. Since  $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$ , the same holds for  $\alpha$ . Compute  $\alpha^2 = 5 + 2\sqrt{6}$ , then  $(\alpha^2 - 5)^2 = 24$ , so  $\alpha^4 - 10\alpha^2 + 1 = 0$ . This polynomial is irreducible (degree 4), so it is the minimal polynomial.

**Problem 5.** Let  $K/F$  be algebraic and  $R$  a ring with  $F \subseteq R \subseteq K$ . Show that  $R$  is a subfield of  $K$ .

*Proof.* For nonzero  $r \in R$ , since  $r$  is algebraic over  $F$ , there exists  $a_n r^n + \cdots + a_1 r + a_0 = 0$  with  $a_0 \neq 0$ . Then  $r(a_n r^{n-1} + \cdots + a_1) = -a_0$ , so  $r^{-1} = -\frac{1}{a_0}(a_n r^{n-1} + \cdots + a_1) \in R$ .

**Problem 6.** Prove that if  $K/F$  is algebraic and  $\sigma : K \rightarrow K$  is an  $F$ -homomorphism, then  $\sigma$  is an isomorphism.

*Proof.*  $\sigma$  is injective. For surjectivity, let  $\alpha \in K$  and let  $p(x)$  be its minimal polynomial over  $F$ . The set  $S$  of roots of  $p$  in  $K$  is finite.  $\sigma$  maps  $S$  into  $S$  injectively, hence bijectively. Thus  $\alpha \in S = \sigma(S)$ , so  $\alpha = \sigma(\beta)$  for some  $\beta \in K$ .

**Problem 7.** If  $F \subseteq K \subseteq L$  and  $K/F$  and  $L/K$  are normal, is  $L/F$  normal? Give a counterexample.

*Solution.* Not in general. Example:  $F = \mathbb{Q}$ ,  $K = \mathbb{Q}(\sqrt{2})$ ,  $L = \mathbb{Q}(\sqrt[4]{2})$ . Then  $K/F$  is normal (splitting field of  $x^2 - 2$ ),  $L/K$  is normal (degree 2, splitting field of  $x^2 - \sqrt{2}$ ), but  $L/F$  is not normal because the minimal polynomial of  $\sqrt[4]{2}$  over  $\mathbb{Q}$  is  $x^4 - 2$  which has a root  $i\sqrt[4]{2} \notin L \subset \mathbb{R}$ .

**Problem 8.** Let  $K, L$  be intermediate fields of  $E/F$ .

1. If  $K/F$  is normal, show that  $KL/L$  is normal.
2. If both  $K/F$  and  $L/F$  are normal, show that  $KL/F$  and  $K \cap L/F$  are normal.

*Proof.*

1.  $K$  is a splitting field over  $F$  of a family of polynomials. These polynomials also lie in  $L[x]$ , so their splitting field over  $L$  is contained in  $KL$ . Hence  $KL$  is normal over  $L$ .
2.  $KL$  is generated by all roots of the polynomials defining  $K$  and  $L$ ; thus it is a splitting field over  $F$ . For the intersection: if an irreducible  $p(x) \in F[x]$  has a root in  $K \cap L$ , then because  $K/F$  and  $L/F$  are normal, all its roots lie in both  $K$  and  $L$ , hence in  $K \cap L$ .

**Problem 9.** Let  $q = p^n$ . Prove that  $x^{q-1} - 1 = \prod_{\alpha \in \mathbb{F}_q^*} (x - \alpha)$  and  $\prod_{\alpha \in \mathbb{F}_q^*} \alpha = (-1)^{p^n}$ .

*Proof.*  $\mathbb{F}_q^*$  is cyclic of order  $q - 1$ ; each element is a root of  $x^{q-1} - 1$ . The polynomial is monic of degree  $q - 1$  and has all  $q - 1$  distinct roots, giving the factorisation.

Comparing constant terms:

$-1 = \prod_{\alpha} (-\alpha) = (-1)^{q-1} \prod \alpha$ , so  $\prod \alpha = (-1)^q = (-1)^{p^n}$ .

**Problem 10.** Decide whether  $i$  belongs to:

1.  $\mathbb{Q}(\sqrt{-2})$
2.  $\mathbb{Q}(\sqrt[4]{-2})$
3.  $\mathbb{Q}(\alpha)$  where  $\alpha^3 + \alpha + 1 = 0$ .

*Solution.*

1. No. If  $i = a + b\sqrt{-2}$  with rational  $a, b$ , squaring gives  $-1 = a^2 - 2b^2 + 2ab\sqrt{-2}$ ; then  $ab = 0$  and  $a^2 - 2b^2 = -1$  leads to contradictions.
2. No. Let  $K = \mathbb{Q}(\sqrt[4]{-2})$ . If  $i \in K$ , then  $\sqrt{2} = i\sqrt{-2} = i \cdot (\sqrt[4]{-2})^2 \in K$ . Then  $\mathbb{Q}(i, \sqrt{2}) \subseteq K$ . But  $[\mathbb{Q}(i, \sqrt{2}) : \mathbb{Q}] = 4$  and  $[K : \mathbb{Q}] = 4$ , so  $K = \mathbb{Q}(i, \sqrt{2})$ . However,  $\sqrt[4]{-2} = 2^{-1/4}(1 + i)$  would then imply  $2^{-1/4} \in \mathbb{Q}(\sqrt{2})$ , impossible.
3. No. If  $i \in K$ , then  $\mathbb{Q}(i) \subseteq K$ . But  $[\mathbb{Q}(i) : \mathbb{Q}] = 2$  does not divide  $[K : \mathbb{Q}] = 3$ , contradiction.

**Problem 11.** Let  $F$  be a field of characteristic  $p > 0$ ,

$a \in F \setminus F^p$ . Prove that  $x^{p^e} - a$  is irreducible over  $F$  for any  $e \geq 1$ .

*Proof.* Let  $\alpha$  be a root in a splitting field; then  $x^{p^e} - a = (x - \alpha)^{p^e}$ . Suppose  $g(x) = (x - \alpha)^d$  is a monic irreducible factor over  $F$  with  $d = mp^k$ ,  $p \nmid m$ . Then  $g(x) = ((x - \alpha)^{p^k})^m = (x^{p^k} - \alpha^{p^k})^m$ . The coefficient of  $x^{(m-1)p^k}$  is  $-m\alpha^{p^k} \in F$ , so  $\alpha^{p^k} \in F$ . If  $k < e$ , then  $a = \alpha^{p^e} = (\alpha^{p^k})^{p^{e-k}} \in F^p$ , contradiction. Hence  $k \geq e$ , and because  $d \leq p^e$  we must have  $k = e$  and  $m = 1$ . Thus  $g = f$ .

**Problem 12.** Prove that a normal closure of a finite separable extension is separable.

*Proof.* Let  $K/F$  be finite separable. Then  $K = F(\alpha_1, \dots, \alpha_n)$  with each  $\alpha_i$  separable over  $F$ . Their minimal polynomials are separable. The normal closure is the splitting field of the product of these polynomials; it is generated by all their roots, which are separable. Hence the normal closure is separable over  $F$ .

**Problem 13.** Let  $K/E/F$  be algebraic extensions.

1. If  $u \in K$  is separable over  $F$ , then  $u$  is separable over  $E$ .
2. If  $K/F$  is separable, then  $K/E$  and  $E/F$  are separable.

*Proof.*

1. The minimal polynomial of  $u$  over  $E$  divides its minimal polynomial over  $F$ ; the latter has no repeated roots, so the former also has none.
2. Every element of  $K$  is separable over  $F$ , hence over  $E$  by (1). For any  $x \in E$ ,  $x \in K$  so  $x$  is separable over  $F$ .

**Problem 14.** Let  $F$  have characteristic  $p > 0$ .

1. If  $f(x) \in F[x]$  is irreducible and  $\deg f$  is not divisible by  $p$ , then  $f$  is separable.
2. If  $[K : F]$  is not divisible by  $p$ , then  $K/F$  is separable.

*Proof.*

1. If  $f$  were inseparable, then in characteristic  $p$  we would have  $f(x) = g(x^p)$ , so  $p \mid \deg f$ , contradiction.
2. For any  $\alpha \in K$ ,  $\deg m_{\alpha}(x)$  divides  $[K : F]$ , so it is not divisible by  $p$ ; by (1),  $m_{\alpha}$  is separable.

**Problem 15.** Give an example of a degree-2 extension that is not separable.

*Solution.* Let  $F = \mathbb{F}_2(t)$ , the field of rational functions over  $\mathbb{F}_2$ . The polynomial  $x^2 - t \in F[x]$  is irreducible (Eisenstein with prime  $t$ ). In characteristic 2,  $x^2 - t = (x - \sqrt{t})^2$  in a splitting field, so the extension  $F(\sqrt{t})/F$  is purely inseparable of degree 2.